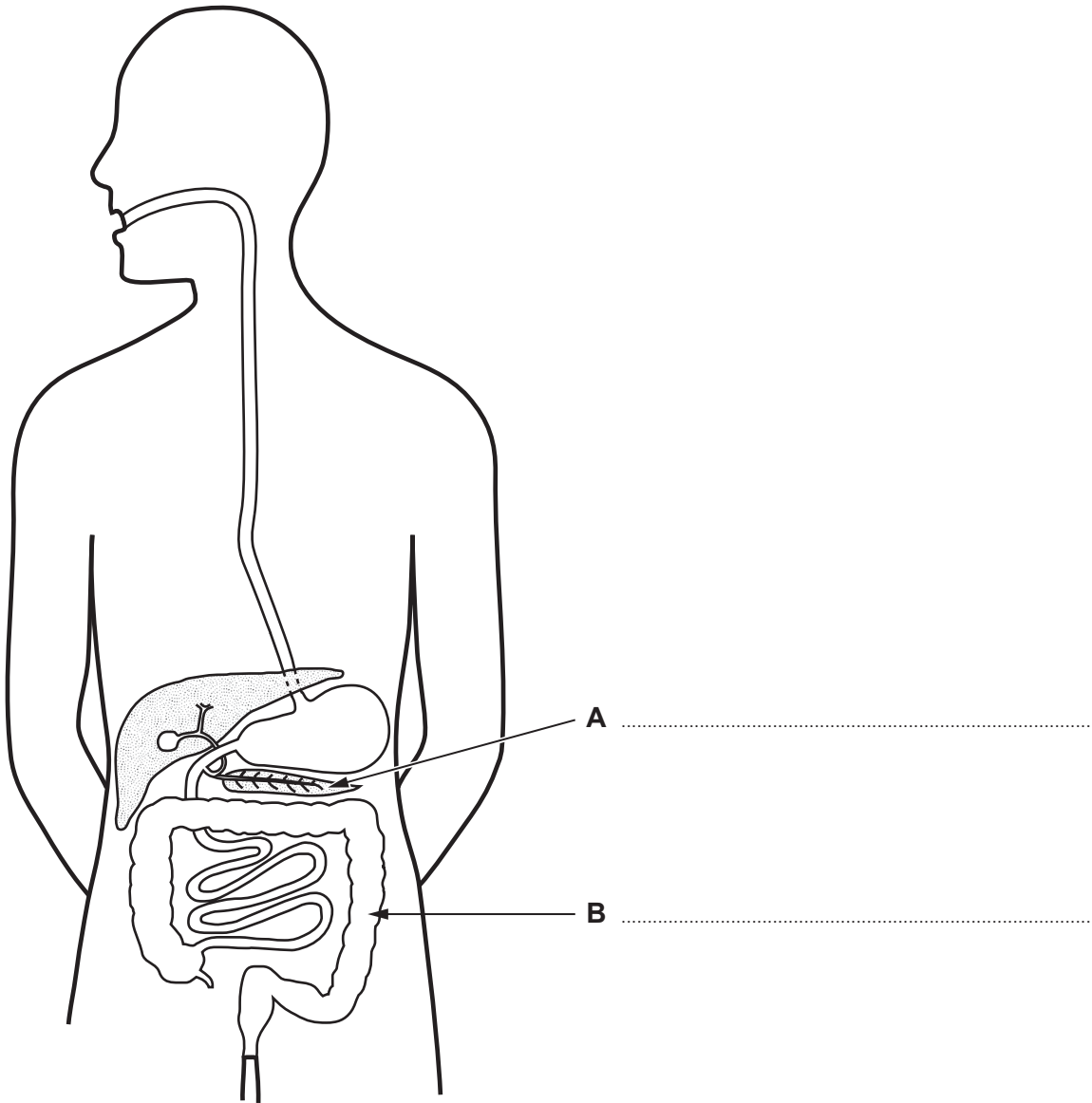


3. **Image 3.1** shows the human digestive system.

Image 3.1



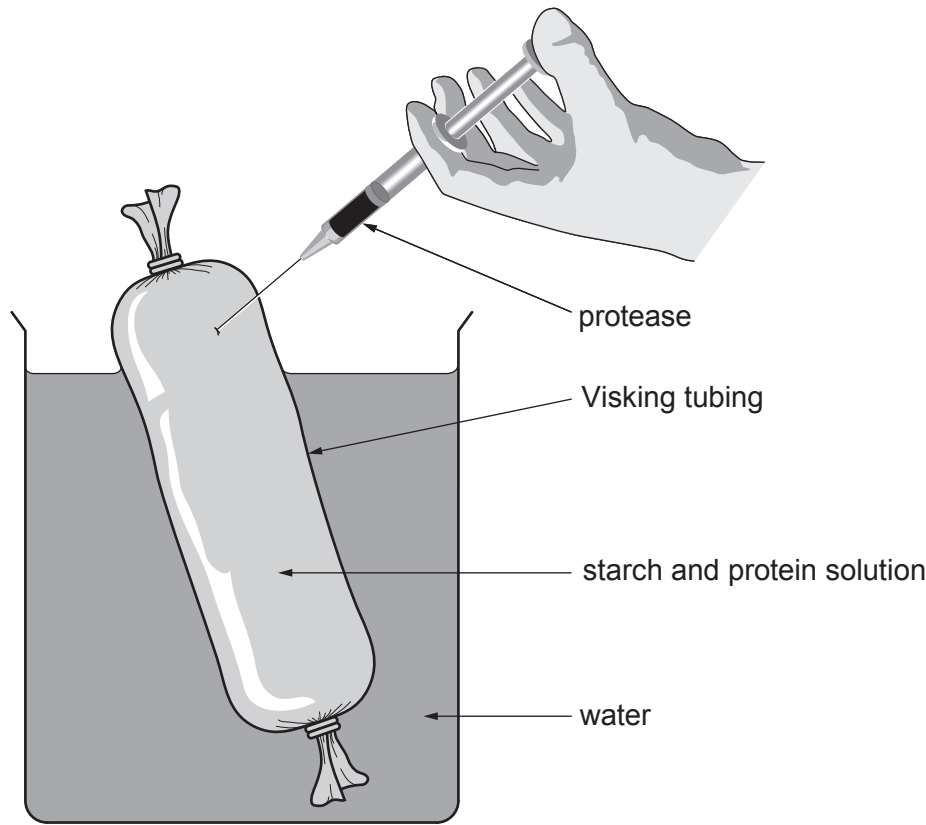
(a) Label parts **A** and **B** on **Image 3.1**.

[2]



- (b) A length of Visking tubing was used to model a part of the digestive system. The Visking tubing was filled with a solution of starch and protein. The apparatus was kept at a temperature of 35°C. The apparatus is shown in **Image 3.2**.

Image 3.2



A sample of the water surrounding the Visking tubing was tested for protein, starch, amino acids and glucose at the start of the experiment. After 15 minutes, protease was injected into the Visking tubing. After 30 minutes, the water surrounding the Visking tubing was tested again.

- (i) Describe how you would use a chemical to test the water for the presence of **glucose**. [2]

.....

.....

.....

.....



Examiner
only

(ii) **Complete Table 3.3** below to show the expected results for this experiment. One column has been done for you.

✓ = present in the water surrounding the Visking tubing
✗ = absent in the water surrounding the Visking tubing

[2]

Table 3.3

Time /minutes	Starch	Protein	Amino acids	Glucose
0	✗			
30	✗			

(c) Suggest which part of the digestive system shown in **Image 3.1** is represented by the Visking tubing. Give a reason for your answer.

[2]

.....

.....

.....

.....

8

3400UA01
13

